An empirical test of the "daily and weekly effect" of China's A-share market

——Take the Shanghai Composite Index and Shanghai Pudong Development Bank as examples

1、 Abstract

The diurnal effect is a widely studied calendar anomaly in financial markets, specifically referring to the phenomenon that the yields of stocks or other financial assets show systematic differences in different transactions during the week. Different stock markets have different daily and weekly effects. For example, in European and American markets, there is a "Monday effect", where the average yield on Monday is significantly negative, which is thought to be related to investor sentiment, information releases, settlement cycles, or risk-averse behavior.

If the daily and weekly effect exists and is stable, it can be considered that the stock market is not fully effective because investors can build their investment strategies based on publicly available information such as dates to obtain excess profits. If the daily and weekly effects are not significant or have basically disappeared, the stock market can be considered effective.

Nowadays, with the maturity of the market, the improvement of trading mechanisms and the change of investor structure, the daily and weekly effects of the stock market have gradually weakened or even disappeared. This report will

use the Shanghai Composite Index and Shanghai Pudong Development Bank as examples to study the daily and weekly effects of China's A-share market.

2、 Study design and data

This study aims to test whether there is a "day-to-week effect" in China's stock market, that is, whether there is a systematic difference in stock yields between different trading days. In order to make the research true and effective, we should choose stocks that can represent the overall market and have good liquidity, so I chose the daily closing price of the Shanghai Composite Index and Shanghai Pudong Development Bank, with a sample range of 2025.11.27-2026.3.25, a total of 77 trading days, and a sample capacity of 76. The data comes from the website http://q.stock.sohu.com/zs/600160/lshq.shtml. Each trading day was classified from Monday to Friday, and the sample size, mean and variance of each group were counted separately.

First, define the yield formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Among them, R_t is the return on day t , P_t is the closing price on day t , and P_{t-1} is the closing price on the previous trading day.

Then, use SPSS software to do basic statistics on the yields of the two stocks. The average and variance of the daily yield of the Shanghai Composite Index from

Monday to Friday is shown in Table 1.

Table 1 Basic statistics of the return of the Shanghai Composite Index by day

week	Sample size	Average	Difference
1	14	-0.005	0.000025
2	16	0.0149	0.000222
3	16	0.0036	1.3E-05
4	15	0.0144	0.000207
5	15	0.0096	9.22E-05

The average and variance of the daily yield of Shanghai Pudong Development Bank from Monday to Friday is shown in Table 2.

Table 2 Basic statistics of SPD Bank's daily yield

week	Sample size	Average	Difference
1	14	0.0032	1.02E-05
2	16	0.0169	0.000286
3	16	0.03549	0.00126
4	15	0.02982	0.000889
5	15	0.02725	0.000743

In order to further test whether the difference in returns on each trading day is significant, the one-way analysis of variance method is used, that is, whether the difference in returns is purely caused by accidental factors. The factor here is

the date, with five levels from Monday to Friday.

The yield on Monday, Tuesday , ⋯ , Friday is R_i ($i = 1,2,3,4,5$), assuming (1); $R_i \sim N(\mu_i, \sigma_i^2)$ (2) each σ_i^2 is equal; (3) The yields of each trading day in a week are independent of each other. Test the hypothesis:

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_5 \qquad H_1: \mu_1, \mu_2, \ldots, \mu_5 \text{不全相等}$$

The variance analysis table of the Shanghai Composite Index for each day is as follows:

Table 3 Analysis of variance of the Shanghai Composite Index

	sum of squares	Degree of freedom	Mean square	F	P-value
Between groups	0.004	4	0.001	0.715	0.585
Within the group	0.101	71	0.001		
Total	0.105	75			

The variance analysis table of SPD Bank's daily yield is as follows:

Table 4 Analysis of variance of SPD Bank

	sum of squares	Degree of freedom	Mean square	F	P-value
Between groups	0.000	4	0.000	0.021	0.999
Within the group	0.053	71	0.001		
Total	0.053	75			

3、 Analysis of results

As shown in Table 1, Monday's average was -0.5%, the only day with negative returns, indicating that the market may be weak at the beginning of the week. Tuesday to Friday are both positive returns. The Shanghai Composite Index exhibits a typical "Monday effect", where Monday is often a falling day while other trading days are generally up. From a volatility perspective, the variance on Tuesday and Thursday fluctuated greatly; Wednesday has the smallest variance, indicating that the price change on that day is relatively stable; Friday was the second, with moderate volatility. Tuesday and Thursday may be active periods of capital flows, leading to increased volatility; Wednesday was calmer, which may be related to the wait-and-see sentiment in the market.

As shown in Table 2, all trading days are positive returns, and no negative returns are the highest on Wednesday. SPD Bank shares showed a strong "mid-week effect", which may be due to major events in bank stocks on Wednesday or the market's "buying on dip" behavior pattern towards the banking sector. In terms of volatility, Wednesday has the largest variance, followed by Thursday and Friday, and Monday is the most stable. Wednesday is not only the most profitable day, but also the most volatile, and there may be drastic changes in market sentiment towards the stock, or there may be information shocks or liquidity risks. Monday is the smoothest, probably because the market is in a state of rest.

From Table 3, $F = 0.715$, $p = 0.585 > \alpha = 0.05$,the statistics cannot reject the null hypothesis H_0 , that is, it cannot be considered that there is a significant

difference in the average return of the Shanghai Composite Index between five trading days a week, so it is believed that the Shanghai Composite Index does not have a significant daily weekly effect.

From Table 4, $\mathrm{F} = 0.021$, $\mathrm{p} = 0.999 > \alpha = 0.05$, We fail to reject the null hypothesis H_0 ; that is, there is no sufficient evidence to suggest a significant difference in the average returns of the SSE Composite Index across the five weekdays. Therefore, we conclude that the SSE Composite Index does not exhibit a statistically significant day-of-the-week effect.

This study preliminarily examines the daily and weekly effects of the Shanghai Composite Index and SPD Bank through descriptive statistics and one-way analysis of variance, but there are still obvious limitations in relying only on mean, variance and P value. For example, Table 2 shows that SPD Bank's average yield on Wednesday was as high as 3.55%, much higher than other trading days, and it intuitively seems that there is a significant "mid-week effect"; However, ANOVA showed that the difference was not statistically significant. This contradiction of "high mean but not significant" is likely to stem from problems such as high skewness of yield distribution, extreme interference or excessive intra-group fluctuations that are not fully reflected by the numerical summary. Therefore, further diagram analysis is made.

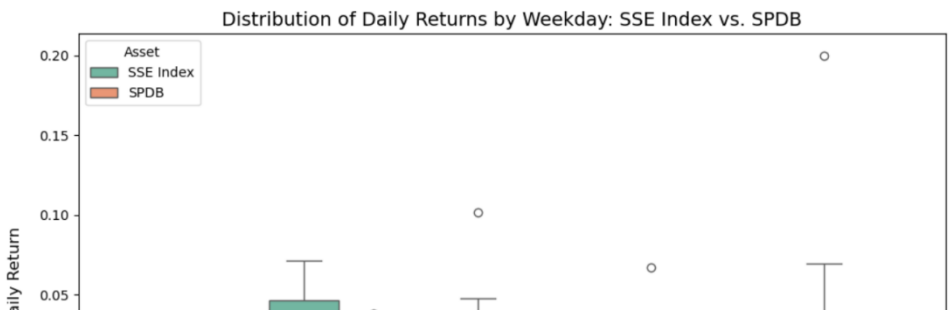

Figure 1 Distribution of the yield of the Shanghai Composite

From the figure, it can be observed that the Shanghai Composite Index (SSE Index) exhibits a trend of rising first, then declining, and subsequently rebounding. Tuesday shows the best performance, although it contains a negative outlier. Thursday is the only day with a negative median return, and overall volatility is relatively low—consistent with the characteristics of large-cap blue-chip stocks.

Pudong Development Bank (SPDB) appears less active than the broader market; its boxplots are generally narrower, indicating smaller intraday fluctuations and greater stability compared to the market index.

As shown in the plot, although the SSE Index displays a notable positive outlier on Wednesday—which pulls up the mean—the majority of observations on that day cluster within the 0–0.02 range, with a median of approximately 0.01. Moreover, high within-group variability and numerous outliers inflate the variance. Under a small sample size, such an elevated mean driven primarily by isolated extreme values lacks robustness, making it difficult for ANOVA to detect statistically significant differences across weekdays. This explains why the ANOVA

results indicate that the observed weekday effects are not statistically significant.

4、 Discussion and limitations

Firstly, the sample time window is short and focused on a specific time, which may not be representative of long-term market behavior. If the daily and weekly effects exist, they may be time-varying and affected by factors such as macroeconomic cycles, policy regulation, and changes in investor structure.

Secondly, the non-normality of the yield distribution may affect the validity of ANOVA. ANOVA requires that each set of data roughly obeys a normal distribution and has homogeneity of variance, but financial returns often show the characteristics of peaks, thick tails, and skews. The box plot shown in the figure also reveals the presence of multiple outliers, which may violate model assumptions and bias the P-value.

Third, the limited sample capacity is one of the main limitations of this study. The full sample contains only 76 valid trading days, and the sub-sample size after each trading day is more numerous. Under small sample conditions, even if there is a real calendar effect, the power of statistical tests is low, and it is difficult to detect moderate or weak effects.

5、 Conclusions and implications

Based on the daily return data of 76 trading days from November 27, 2025

to March 25, 2026, this study empirically tests whether there is a "daily-weekly effect" in the A-share market through descriptive statistics and one-way analysis of variance. The study found that:

(1) The Shanghai Composite Index showed a negative average return on Monday, and the rest of the trading days were positive, which is ostensibly in line with the classic "Monday effect"; Shanghai Pudong Development Bank showed a "mid-week effect", with an average yield of 3.55% on Wednesday, significantly higher than other trading days;

(2) The results of analysis of variance showed that the differences between the two groups did not reach the statistical significance level, and the null hypothesis that "there is no systematic difference in the average return of each trading day" could not be rejected.

(3) During the sample period, China's A-share market did not show a statistically significant daily weekly effect. This knot shows that the market has been able to digest public information related to the trading day relatively quickly, and it is difficult for investors to continue to obtain excess returns based on the simple rule of date alone.

Through this study, the following enlightenment can be obtained:

First, investors should not be blindly superstitious about empirical strategies such as investing according to dates. In the context of the current A-share market system becoming increasingly perfect and the efficiency of information dissemination is improving, simple calendar trading strategies are likely to have

been priced in the market, making it difficult to bring stable excess returns. For regulators and exchanges, the weakening of the daily effect can be seen as a positive sign of market maturity. This reflects that the optimization of information disclosure mechanisms, trading systems and investor structures is driving the evolution of the market to higher efficiency.

Second, for subsequent research, the exploration can be deepened from the following directions: expanding the time span of the sample to cover multiple bull-bear cycles to identify whether the diurnal effect is time-varying or periodic; More representative targets were introduced to test the universality and heterogeneity of the effect.

To sum up, the daily weekly effect, as an important window to test the effectiveness of the market, has guiding significance for investment practice. This study does not find a significant diurnal effect, indicating that China's A-share market is becoming more effective in this dimension, and the traditional calendar anomaly is gradually disappearing, which is consistent with the development trend of major global markets. However, this does not mean the end of research value, but rather suggests the need to look at the microstructure and behavioral logic of the market from a more refined, dynamic and multi-dimensional perspective.